BIOT 6900 Module 7

Nicholas Adrogue

Introduction

The purpose of this module was to take the separated data found and analyzed in the previous modules and put it all into a knowledge graph which can be queried in the future. Graphs are very good at incorporating different bodies of knowledge together in an easily accessible way, with each vertex containing a piece of information and every edge relating them to another. To create this graph, I used Neo4j [1]. For PageRank and community detection, I used NetworkX [2]

Building the Graph

The graph vertices contain 19 genes, 20 patients, 11 phenotypes, 10 drugs, 9 pathways, 9 variants, and 4 diseases. The edges contain 22 gene-disease associations, 18 disease-phenotype relationships, 18 gene-pathway relationships, 11 drug-target links, 9 variant-gene links, and 7 patient-variant connections.

The genes include the 15 top-ranking genes from my module 3 targets and 4 related disease drug targets. In module 3, I used protein language embeddings from [3] and [4] with tractability from [5]. The GWAS data came from MONDO_0004975 [6], the RNA data came from GSE104704 [7], and the protein data came from syn12269876 [8]. The druggability came from the maximum between p_sm and p_ab from the tractability model. The HCN1 druggability comes from the evidence in module 3, and HCN1 has a module 4 pocket score of 0.878, using the PDB 8Y60 structure [9], fpocket detection [10], and AlphaFold models from [11] and [12].

The phenotypes and related diseases came from [13]. Disease IDs came from [14] and [15]. For Orphanet, the data came from ORPHA:51. The pathways came from [16].

The drugs came from Open Targets candidates [5][17][18]. Only 3 approved antibodies made the cut. HCN1's relationship to Ivabradine was not in those databases, since HCN4 is its only listed target [19], so it was added separately based on the research from previous modules [20][9][21].

For the variants, the top GWAS single-nucleotide polymorphisms were taken from [6], and APOE's ε4 and ε2 were added as well.

For the patients, 20 people were chosen from the module 6 group [22], with scoring from PGS000334 [23][24]. The age and case status are simulated.

Every gene uses the HGNC symbol as the key, and uniqueness is asserted on the name, ensuring each gene only gets one node. Ensembl IDs were added from [25].

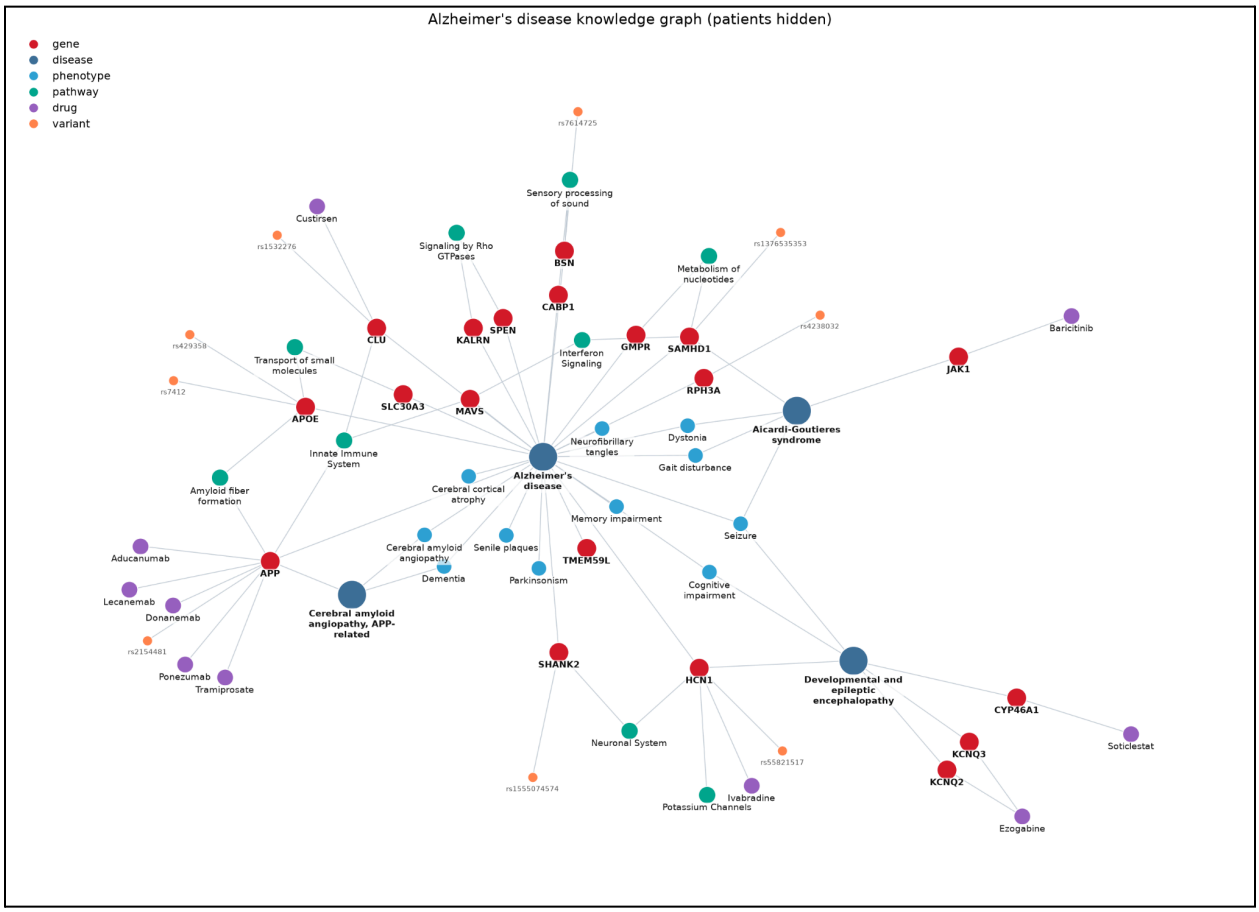


Figure 1 Knowledge graph. The patients are hidden for legibility

Schema and Evidence

The node types are: genes, diseases, phenotypes, pathways, drugs, variants, and patients. Genes serve as the nodes that carry evidence from earlier modules. The disease nodes are split into Alzheimer's Disease (AD), which is the anchor node, and other diseases that serve for repurposing the graph. The phenotype nodes connect diseases, the pathway nodes represent mechanisms that multiple genes might use, drug nodes indicate potential results, variant nodes demonstrate differences in genes that may correlate with disease, and the patient nodes show how this might present in a real person.

The edges are as follows. The ASSOCIATED_WITH tracks gene-disease links to show which genes are associated with which diseases. HAS_PHENOTYPE goes from disease to phenotype to track which diseases lead to which phenotypes. PARTICIPATES_IN links genes to pathways they participate in. TARGETS links drugs to proteins from which gene they target. OCCURS_IN matches variants to the genes they affect. CARRIES links patients to variants they carry.

Each node and edge carry a source property to cite where they originated from. Every node except for the patients has a standard ID, which helps keep consistency. The “ASSOCIATED WITH” edges for Alzheimer's contain the GWAS p-value for confidence. The edges to other related diseases do not carry the p-value, since it is specifically calculated for AD.

Top Targets

The top targets are shown below in Table 1:

Gene	PageRank	GWAS p	Druggability
APOE	0.047	~1e-300	0.016
APP	0.037	2e-14	0.048
HCN1	0.017	4e-13	0.879
CLU	0.017	1e-47	0.128
SAMHD1 / SHANK2 / RPH3A / BSN	0.011	various	various

Table 1 Top targets

APOE ranking first is to be expected. ε4 is a very strong genetic risk factor for AD [26][27]. APP being second is also expected. It is the target of several drugs that treat AD [28][29][30]. CLU is also established as being related to AD [31]. These rank high from PageRank because they have many incoming edges, which is what PageRank uses to rank the nodes [32]. This means that the ranking can be influenced by how “popular” a gene is, that is to say, how much research and application a gene has already undergone in terms of AD.

HCN1 being tied for third is surprising, since it is not a classic AD gene. Many of its connections go instead through epilepsy [33] and ivabradine. It is linked to AD through the hyperexcitability of neurons, as seen in [34]. The third strongest GWAS p value was for SAMHD1 with 2e-36. It has a druggability score of 0.83.

Hypothesis from a path

One path that is interesting is (Ezogabine) -[:TARGETS]-> (KCNQ2) -[:ASSOCIATED_WITH]-> (Developmental and epileptic encephalopathy) -[:HAS_PHENOTYPE]-> (Seizure) <-[:HAS_PHENOTYPE]- (Alzheimer's disease). Each link in this path has its own evidence: KCNQ2 to developmental and epileptic encephalopathy (DEE) is a Mendelian link [13][14] and Ezogabine to KCNQ2 comes from [5]. This leads to a hypothesis: Some AD patients have network hyperexcitability, and drugs that can lower neuronal activity via potassium and/or HCN channels could slow cognitive decline in this group.

There is supporting evidence from the literature. For example, epileptiform activity can occur in early-stage Alzheimer's and is linked to faster cognitive decline [34]. An anti-seizure drug named levetiracetam improved cognitive functions in AD patients with epileptic activity [35]. Additionally, ezogabine itself is a KCNQ-related drug [36]. To test this, expression tests would need to be done for HCN1, KCNQ2, and KCNQ3 in AD patients vs a control group. An additional test would be to use animal models to examine the effect in a mouse with AD. Ezogabine has been taken off the market, so it would have to be examined as a lead compound, not the drug itself.

The communities are greedy clusters [37]. One community to look at is the group including HCN1, SHANK2, Ivabradine, Neuronal System, Potassium Channels, and their lead SNPs. This community seems to be a network hyperexcitability grouping. This cluster further supports the hypothesis.

Edges being evidence

The weakest edges from my graph are JAK1 → AGS, CYP46A1 → DEE, and KCNQ3 → DEE. These are association edges, and only exist because a certain drug for the given target affects the given gene. KCNQ3's evidence for relating to AD is weak. Its AD GWAS p value is 6e-6, which is not significant compared to the cutoff value of 5e-8 [38]. Some phenotypes are only connected because one term is the parent of another [13]. All of this is to say, edges being present show evidence for connectedness, but do not always ensure true correlation between nodes. AI was used to help pull data from earlier modules and gather more data including HPO, Reactome, and Open Targets data, organizing it into a cohesive graph. I checked the sources, including the gene values from my previous modules, IDs from databases, and citations against Crossref and PubMed. Real phenotype data would improve on the understanding of real-world impact.

Sources

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